

Novel Human Interface Device: Magic Wand

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ABSTRACT

This paper investigates how an innovative human-interface device can help enchant everyday household routines. A prototype called the “Magic Wand” was developed, capable of controlling common home devices such as lights and music speakers. To make the user experience more magical, a gesture-based control scheme was implemented, allowing users to perform a “spell” gesture and point the wand at a desired device to control it.

The development process involved several iterations and incorporated user feedback throughout. The final design prioritized ease of use, a sense of enchantment, and modularity. In response to user input, the design also included haptic and visual feedback, an improved and more comfortable grip, and simplified control gestures.

Although the Magic Wand did not result in significantly higher enchantment scores compared to a manually controlled lamp, users still reported higher levels of perceived enchantment. This suggests its potential in transforming ordinary household routines into magical experiences.

1 INTRODUCTION

Everyday household routines, such as turning on lights or the radio, often involve manual and repetitive actions. To solve this inconvenience, smart home devices are becoming more popular to interact with these systems [3]. The adoption of these automation systems in households worldwide increases daily, because of benefits such as energy efficiency; convenience and resource management. We define automation as a technique of controlling a process through electronic devices while minimizing human involvement [4]. While automation enhances efficiency, the user experience plays a crucial role in ensuring that these systems are not only functional but also engaging; intuitive and even enchanting [6]. There are multiple interpretations of the term enchantment: Bennett refers to the experience of it as “both caught up and carried away” [1].

We research how a human-interface device can be designed to enchant household routines through gesture-based interactions while maintaining ease of use and user satisfaction. To explore this, we developed multiple low-fidelity prototypes to test our initial concept and interactions with the prototype, which took on the shape of a *Magic Wand*. After that, several sketches were made for the high-fidelity prototype and the final design was modeled in Autodesk Fusion. The prototype was 3D-printed and programmed for gesture-based control. Through user testing and evaluation, we examined how the device design influenced usability; engagement and perceived enchantment. This paper contributes to the broader research on enchanting technologies. Beyond improving household

routines, we highlight new ways to enhance the user experience and create more meaningful and fun connections between users and technology.

The *Related Work* section discusses existing literature and identifies research gaps, and concludes with the research questions of this paper. To answer these questions, a *Magic Wand* was designed and evaluated. The *Design* section outlines the design process and how we came to the final design, followed by the *Implementation*, where all technical details are described. The *Evaluation* section describes the setup of the user study, while the *Results* elaborate on these findings. We continue with discussing the highlights to our study’s; the limitations and opportunities, providing potential directions for future research. Finally, key findings are summarized and interpreted in the conclusion of this study.

2 RELATED WORK

In human-computer interaction, enchantment has been explored as an alternative perspective on user experience [6]. Enchantment is defined as a state of being caught up and carried away while experiencing a sense of wonder and unexpectedness. Therefore, McCarthy et al. argue that designing for enchantment involves sensibilities such as sensory engagement, emotional connection, playfulness, ambiguity, and transformation. These elements stimulate both thought and feeling and thus mostly emphasise affective qualities of interaction. Sengers et al. (2008) [9] discuss the importance of emotional experience in computing design. They suggest that enchantment emerges from designing for open-ended engagement and user interpretation, and not from direct engineering.

In the context of smart home interfaces, previous studies have focused on various aspects of improving user experience. Zhao et al. (2016) [11] recommended using traditional usability metrics – efficiency, effectiveness and accessibility – to improve interaction design for smart home devices. While these usability metrics mainly focus on functionality and practicality, other research has explored more engaging and emotional approaches to smart home interaction. Chamunorwa et al. (2019) [2] found that embedding control interfaces in everyday household objects can improve user experience by bringing playfulness to controlling household devices. More recently, Xu et al. (2024) [10] proposed a design model for smart home technology, specifically for automation. They identified nine factors that positively influence user experience, including emotion, immersion, and aesthetics – elements that align closely with the concept of enchantment. Moreover, their findings indicate that while some factors contribute primarily to the practicality of automation, others play a greater role in enhancing the interaction

experience. This emphasises the balance between functionality and interaction in smart home design is important to consider as well.

Despite these contributions, no concrete research has examined how smart home interactions can be designed to evoke enchantment. This study addresses this gap by investigating how interaction technology can create an enchanting experience when controlling household devices. Moreover, we will examine how enchantment intersects with usability to ensure a balance between enchantment and user-friendliness. Thus, the following research questions will be answered:

- (1) How can controlling devices around the house with a novel remote control – the *Magic Wand* – be made user-friendly?
- (2) How enchanting is it for users to control household devices with a novel remote control, the *Magic Wand*?

Building on Zhao et al. (2016) [11], the *Magic Wand* is designed to streamline the user's interaction with household devices. Efficiency is measured by the speed and ease with which users can execute desired actions, while effectiveness is gauged by the accuracy and reliability of those actions. By grounding our design in these metrics, we ensure that the interface is not only innovative but also practically useful in everyday scenarios.

Accessibility is also paramount in our research, similarly to its appearance in Zhao et al. (2016) [11]. Much of our research focuses on creating a user-friendly experience. Not only that, but user intuition is also taken into account significantly; we will observe how users interact with our prototypes by intuition. This deliberate integration of Zhao et al.'s metrics not only validates the technical merits of our prototype, but also proves its adaptability to a real production-scale device. This last part however, goes beyond the scope of the research.

Building on the work of Chamunorwa et al. (2019) [2], our research hypothesizes that if users find controlling household devices with the *Magic Wand* to be more enchanting, then this effect supports the idea that embedding control interfaces directly into everyday devices can enhance the overall user experience. Chamunorwa et al. (2019) [2] argue that such integration elevates the user experience by introducing a sense of playfulness, making interactions not only functional but also engaging and enjoyable.

Playfulness in this context transforms routine tasks into interactive, enchanting experiences, potentially fostering a more emotional reaction than with a "normal" device. Then, when users are drawn to the device because it offers a novel and enchanting way to manage household functions, it creates a positive feedback loop that reinforces both usability and satisfaction - reinforcing routine behavior in a non-intrusive and fun way.

The novel *Magic Wand* design exemplifies a holistic approach to the design of a remote control human interface device. It challenges the existing standard which merely focuses on efficiency and task execution; by also seeking to elevate everyday interactions by making them more accessible and playfully engaging. Our research is built on the premise that by aligning with established usability metrics while also embracing the concept of playful interaction, the *Magic Wand* can redefine the user experience in the smart home domain. The right balance between usability and enchantment is a key factor in this research.



Figure 1: Low-Fi PTT 1.0

3 DESIGN

3.1 Lo-Fi PTT 1.0

The design process started with creating a low-fidelity prototype to assess user interactions and expectations with the interactive wand, aimed at helping to improve the design and functionalities for further prototypes. The prototype, built by LEGO-bricks (figure 1), had no technical functionality; it was purely a physical object designed to study user interaction. It was shaped to fit comfortably in the hand, allowing us to observe how users naturally hold and gesture with it. A "stand" for quick pick-up was included, as we cared to observe whether users would actually use it or if they would simply toss the wand down post-use. The stand could also help prevent the wand from rolling off surfaces, but whether this was effective ultimately depended on user behavior. Functional interactions were simulated using a phone-flashlight controlled via a smartwatch.

To gather comprehensive feedback, we conducted a structured user research test involving three participants. Both qualitative observations and quantitative data were collected, utilizing the User Engagement Scale (UES) [8] to measure factors such as focused attention, perceived usability, aesthetic appeal, and reward. The protocol of the user research can be found in the appendix, and is attached to submission.

The user testing of this prototype's iteration was especially aimed to investigate:

- How users behave when interacting with the wand;
- Which action-flow (order-of-execution) feels the most enchanting;
- Expected and preferred device-to-user feedback;
- Preferred form factor of the wand in a range of scenarios.

3.1.1 User Feedback. The participants did not really lose themselves in the experience. This is understandable, because this was the first user test with a low-fidelity prototype which also lacked functionality. The limited interactivity may have made it harder for users to engage with it properly. All users reported low to moderate frustration, suggesting the wand did not feel overly complex to use. Some confusion and cognitive effort were noted, which again was expected for this iteration of the prototype. Despite the limited



Figure 2: Low-Fi PTT 2.0

functionality, participants found the concept of the device rewarding and felt that using it was worthwhile. This positive response suggests that the core idea has value and resonates with its users.

Functionality

- Most subjects intuitively gesture, then point;
- Vibrations, sounds, and lights are not off-putting and some users even appreciate them;
- Most intuitive gestures involve a spinning motion;
- Users intuitively repeat gestures and pointing until the expected operation executes.

Design

- Subjects expect Harry Potter-like aesthetics, old and worn in appearance;
- The wand has an expected length similar to a remote control;
- The wand should be lighter than a full water bottle, with weight distributed closer to the user's hand;
- It should be intuitive that the wand is meant to be held at the base of the shaft;
- The material should not be slippery, or the wand should feature a stable grip;
- The wand should be easy to pick up and casually place down on a table, and its design should prevent it from rolling off;
- There should be no element resembling a button, as users will instinctively attempt to press it.

Overall, participants saw potential in the wand's usefulness and impact. However, to create a more immersive and seamless experience, further iterations are needed.

3.2 Lo-Fi PTT 2.0

Informed by this feedback, we developed a second prototype using 3D printing, focusing on the wand's ergonomics and aesthetic qualities (see figure 2). Another user research test with three new participants was performed. The protocol was updated to perform extra tasks, such as turning on music and also turning off the light/-music (see appendix). Although this version was slightly over-sized and did not adhere to our beauty-guidelines, it offered valuable data on user comfort and design preferences.

3.2.1 User Feedback. An important conclusion of this session was that we noted how there is a direct conflict between potential

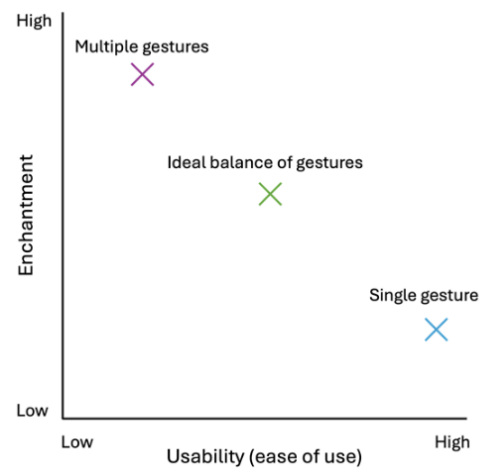


Figure 3: Balancing Enchantment and Usability

enchantment and usability (ease of use). Users mention that implementing multiple gestures for different devices might make interacting with the wand more difficult. However, implementing one gesture for every interaction might make the experience less enchanting for the user. Therefore, a balance between these two factors must be sought, see figure 3.

Functionality

- Distinct gestures improve enchantment, but too many make the device significantly harder to use;
- Opposite directions for turning on and off devices should be explored - simple and intuitive "opposite" gestures;
- Split gesture activation (gesture) and device-targeting (pointing) as two clearly-separated operations;
- Sound, light, and haptic feedback would be nice.

Design

- Use wood to improve the enchantment factor;
- The size is okay, though the size of the wand should be changed;
- Include a (pointy) tip that lights up while doing a gesture.

3.3 Hi-Fi PTT 4.0

After the low-fidelity phase, we started working on the high-fidelity prototype based on the user feedback. We created various options for the handle (lower-part) and staff (higher-part), modeled after the required space for the technical circuit. The sketches are based on the Human-Centered Design Theory, which is a theory that prioritizes the needs, preferences, and limitations of users throughout the design process. This theory supports the notion that products should be designed with a deep understanding of user interaction, which inherently includes considerations of form, function, and comfort, particularly through ergonomics [7]. For the wand, it is important that it performs smoothly and is in line with user expectations. We have gone through various iterations of the high-fidelity prototype, but will only document our final version (figures 4 and 5) considering that the changes are slight, and due to paginal constraints.



Figure 4: High-Fi PTT 4.0 Sep.



Figure 5: High-Fi PTT 4.0 In-Hand

3.3.1 Handle. For the internal technical components to work, the wand must be held in a specific way. Therefore, the handle was designed to be comfortable only when held correctly. In some of our variations, we went as far as placing indentations on the side of the wand to force the fingers into the right position, helping to use the wand correctly.

The *Affordance Theory* plays a big part in how the handgrip influences user interaction and ease of use. This theory focuses on how users perceive the possible actions with a product based on its design. This theory is critical in product design as it explains how the form and function of a product suggest its usage, which directly impacts its ergonomic efficiency and comfort [7]. The chosen design also has a wider top and bottom, which makes it easier to hold and gives it a balanced feel. This shape helps users grip the wand correctly while making it look more like a real magical artifact.

Other designs for the handgrip were not chosen because the rounded bulges might not be comfortable to hold and force fingers into specific positions that do not fit all users. The cutout for the fingers should also not be too straight, because that may not feel as natural in the hand. Eventually, the final design for the handle emerged from sketches B and F (figure 6).

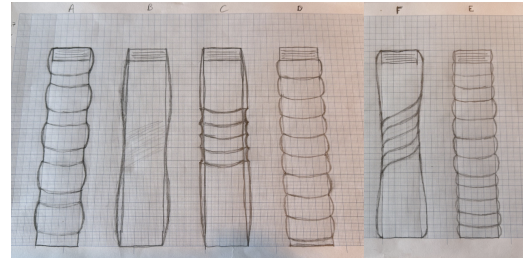


Figure 6: Various Handle-Designs

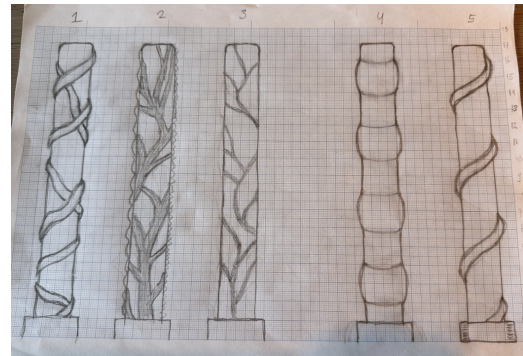


Figure 7: Various Staff-Designs

3.3.2 Staff. The staff design brings an even greater sense of enchantment to the interactive experience. It features a flowing, organic aesthetic that enhances the magical feel. Grounded in the user feedback, the spiral details help the "old-woody" look to match user expectations for a magical device. The design reinforces the immersive and intuitive nature of the interaction. Other designs for the staff were not chosen because the vine-like details alone were not enough to create a magical feel. Other designs felt too bulky, or have spirals that are too perfect. Ultimately, sketch 5 (figure 7) was the best choice for the staff.

3.3.3 Threads. To ensure easy assembly and accessibility, we have implemented a thread system that screws the handle and staff together. This design choice enables easy installation of internal components and also making future adjustments simple and efficient.

3.3.4 Other Aesthetics. We used a wood cutter to add texture to the wand, making it look like real wood. These details give it a natural and handmade feel. The brown color further reinforces this wood-imitation. This design was based on user feedback, as people expected a traditional wooden design and feel for a magical wand. Together, the texture and color make the wand more realistic and enchanting to use.

3.3.5 Gestures. The wand's gestures are designed to be simple, intuitive, and magical. A *downward-tap* motion turns the light on or off, because this felt most natural to the participants of the user test. *Swiping* the wand left or right allows users to change the lamp's color or cycle between different modes. In an earlier iteration, a *double tap* served as an undo function, letting users quickly reverse

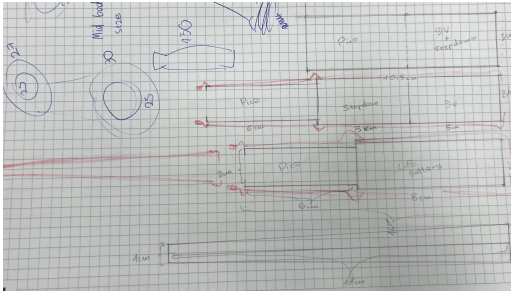


Figure 8: Dimensionality Sketches of Modular-Design

their last action. However, due to confusion with test-users, this feature has been scrapped. The current gestures were chosen based on user feedback, ensuring they feel natural while maintaining the enchanting experience of using a magic wand.

User-testing also showed that some users appreciated vibrations or lights when the correct movement was detected. So, we have added device-to-user feedback features to make interactions more engaging while keeping the wand easy to use.

3.4 Modularity

An essential part of our design was the inherently modular design of our staff. Comfortable positioning or design in a handle is subject to opinion. For instance, left-handed people would find a device designed for right-handed use less comfortable. In an attempt to accommodate the largest part of the population, we have decided to proceed with modular design principles. A positive side-effect of this accessibility feature is that design-preferences in terms of customization can also be personalized, further enhancing the user experience. Furthermore, it allows for easier parallel-working between prototyping and design, as the prototyper can rest assured that if the circuit works with the inner-shell, it will work with any outer-shell the designer creates.

We have achieved modularity by splitting the handle of the Magic Wand with the staff, using a thread. With the thread being standardized across our design principles, we can mix and match any combination of handle- and staff. This would in theory, allow us to 3D-print personalized handles and staffs, and switch swiftly between them.

For this to work seamlessly, and to be less prone to error, of course some standardizations have to be set. With this in mind, we have come up with the following concept for both the handle and the staff: the differentiation between the inner-shell and the outer-shell of the design (as seen in figure 8). The inner-shell considers only the minimum required dimensions, and is in no way beautified or visible to the user. The minimum dimensions are determined by the internal circuit and space required, and will generally be as small as physically possible while still being functional. The outer-shell is visible to the user and must be beautified. It is also exactly the part that can be customized. Hence, a different handle (for left-handed people for instance), would be built on the outer-shell of the device. The only consideration with designing the outer-shell is that it may not violate the constraints set by the inner-shell.

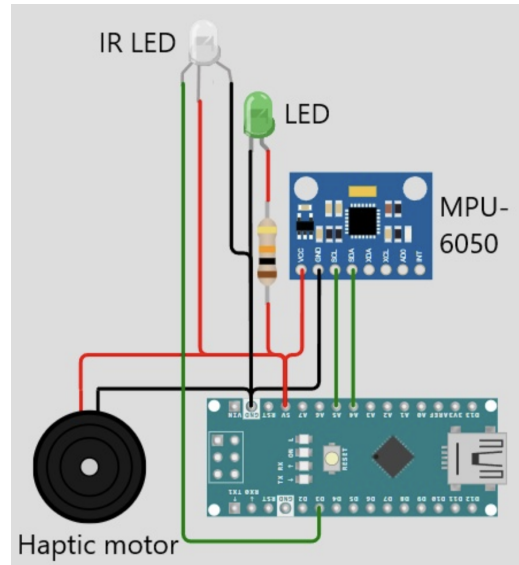


Figure 9: Illustrative Circuit of Primary Components

In order to ease manufacturability, we print the inner-shell and the outer-shell, merged. This means that they are not separable from each other. Though, considering the thread is part of the inner-shell and therefore constrained, this means that any outer-shell customization, whether that is in the handle or in the staff, will inherently fit with the other part of the device.

4 IMPLEMENTATION

4.1 Transmitter

4.1.1 Formalities. Table 1 formally lists all of the parts used for the internal mechanism of the transmitting device.

Table 2 formally lists all of the depending libraries used for the implementation of the device. The source code of these libraries is linked as well. This list is not exhaustive, as the more-common libraries have been left out for clarity purposes.

Figure 9 shows the primary components of the internal circuit for the device. This figure leaves out trivial connections, e.g. the battery to the charging module, step-up converted, et cetera.

4.1.2 Code. The computational logic for the transmitting device is explained below. A variety of gestures can be recognized and each one will send a different signal to the receiver. The main loop and one of the gesture-functions are shown below using pseudo-code for demonstration purposes.

```

1 void loop() {
2     sensors_event_t a, g, temp;
3     mpu.getEvent(&a, &g, &temp);
4
5     float xAccel = a.acceleration.x;
6     float yAccel = a.acceleration.y;
7     float zAccel = a.acceleration.z;
8
9     unsigned long currentTime = millis();
10
11     xLeftFlick(currentTime, xAccel);
12     xRightFlick(currentTime, xAccel);
13     singleTap(currentTime, zAccel, yAccel);

```

Part	Purpose
Arduino Nano	Controlling computational logic.
MPU-6050 sensor	Read the accelerometer to recognize gestures.
Infrared transmitter	Sending infrared signals.
LED light	Provide users with visual feedback on whether a gesture was recognized.
Vibration motor	Provide users with haptic feedback on whether a gesture was recognized.
3V to 5V step up converter	Manages voltage from battery to the board.
Charging module	Allows for battery management and charging.
Rechargeable lipo battery 3.7V	Powers the wand.

Table 1: Formal list of Parts - Transmitter

Library	Purpose
Adafruit_MPU6050.h	Used to read values from the MPU-6050 sensor.
Adafruit_Sensor.h	Unified layer for reading values from Adafruit sensors, like the MPU-6050.
IRremote.hpp	Used to send infra-red signals.
Arduino.h	Access to standard definitions for working with an Arduino Microcontroller.
Wire.h	Used to connect the pins on the Microcontroller to the MPU-6050.

Table 2: Formal list of Libraries - Transmitter

```

14 // Turn off LED after the flick duration
15 if (flickDetected && millis() - flickTime >
16     flickDuration) {
17     flickDetected = false;
18     digitalWrite(ledPin, LOW);
19 }
20 if (currentTime - previousMillis >= interval) {
21     digitalWrite(vibrationMotor, LOW);
22 }
23
24 delay(10); // Small delay to stabilize readings
25 }

```

Listing 1: function for gesture detection

The main loop of the wand reads the *x*, *y* and *z* values from the MPU-6050 accelerometer. Next, it calls three functions: *xLeftFlick*, *xRightFlick*, and *singleTap* which use the accelerometer values as input. Finally, once a gesture has been completed, the LED in the tip of the wand is disabled.

```

1 void xRightFlick(long currentTime, float xAccel) {
2     if (xAccel > 20) {
3         lastDipTime = currentTime; // Store time of dip
4         if (xAccel > 45 && !flickDetected) {
5             if (currentTime - lastDipTime < dipTimeWindow) { //
6                 Ensure dip happened just before flick
7                 flickDetected = true;
8                 flickTime = millis();
9                 digitalWrite(ledPin, HIGH);
10                digitalWrite(vibrationMotor, HIGH);
11                previousMillis = currentTime;
12
13                while (sCounter < sRepeats) {
14                    IrSender.sendNEC(0x00, 0x52, 1); // Send IR
15                    signal
16                    delay(5);
17                    sCounter++;
18                    if (sCounter == 3) {
19                        digitalWrite(vibrationMotor, LOW);
20                    }
21                    Serial.println(sCounter);
22                }
23            }
24        }
25    }
26 }

```

```

21     sCounter = 0;
22 }
23 }
24 }
25 lastXAccel = xAccel; // Update last acceleration value
26 }

```

Listing 2: function for firing action

Three gestures are recognized: a left flick (*xLeftFlick*), a right flick (*xRightFlick*) and a tap of the wand on the hand (*singleTap*). All gesture functions follow the same two phase structure: 1) gesture detection, followed by 2) user feedback and finally 3) sending of an IR-signal. Only the values from the accelerometer and the sent IR-signal vary between the different functions. The code snippet below is for a flick to right, but the logic is the same for all functions. Table 3 shows the values for all functions.

A right flick is signified by an acceleration peak on the *x*-axis (*xAccel* > 20), followed by a spike in the *z*-axis (*zAccel* > 45). Finally, the code checks whether the initial dip occurs shortly after the spike that follows it to confirm that a gesture has been performed (*if (currentTime - lastDipTime < dipTimeWindow)*). If all these conditions are satisfied, the code continues to the user feedback and the infrared signal transmission.

The wand uses a LED in the tip (*digitalWrite(ledPin, HIGH)*) and a vibration motor (*digitalWrite(vibrationMotor, HIGH)*) in the handle for visual and haptic feedback. Both are turned on immediately after the gesture has been detected. The vibration is turned off earlier, as only a short burst suffices to provide the user with feedback. A double vibration pattern has been added to the single tap gesture.

After a gesture has been detected and while the user feedback is being provided, the IR-signal (*IrSender.sendNEC(0x00, 0x52, 1)*) is sent 10 times (*sRepeats*) by means of a counter (*sCounter*) inside a while-loop. Every time an IR-signal is sent, the wand pauses for 0.05 seconds, so that the wand does not send the IR-signal as

Function	Initial Peak/Dip	Gesture Condition	IR Signal Command
xLeftFlick	xAccel < -20	xAccel < -45	0x42
xRightFlick	xAccel > 20	xAccel > 45	0x52
singleTap	zAccel > 25	zAccel > 30 && yAccel > -5	0x64

Table 3: Formal list of Gesture-Values

one continuous signal, which would be illegible for the IR-receiver. Sending various separate signals is required, because users often do not finish their gesture pointing exactly at the IR-receiver, so these extra signals provide them the opportunity to fix their aim.

4.2 Receiver

4.2.1 Formalities. Table 4 formally lists all of the parts used for the internal mechanism of the receiving device.

Table 2 formally lists all of the depending libraries used for the implementation of the device. The source code of these libraries is linked as well. This list is not exhaustive, as the more-common libraries have been left out for clarity purposes.

4.2.2 Code. The main loop decodes the received signal. Then proceeds to check if the received command is a recognized command, after which the corresponding action is taken by a function call. The specific functions will be explained below.

```

1 void loop() {
2   currentTime = millis();
3
4   if (currentTime - lastReceivedTime > 2000) {
5     lastCommand = 0xFF; // Set to an invalid command
6   }
7
8   if (IrReceiver.decode()) {
9     /*
10    * Check the received data and perform actions
11    * according to the received command
12    */
13     uint8_t command = IrReceiver.decodedIRData.command;
14
15     if (command != lastCommand) {
16       lastReceivedTime = currentTime;
17       lastCommand = command; // Avoid duplicate triggers
18
19       switch (command) {
20         case (...):
21           // Action code here
22           break;
23         default:
24           Serial.println("non-existent_command");
25           Serial.println(command);
26           break;
27       }
28     }
29   }
30 }

```

Listing 3: function for IR-controlled receiver

Currently, the receiver is configured to recognize 4 commands. Command *0x10* is the command used to turn on in this case the LED. The *colorSelector* function makes sure the RGB LED is turned on using the color that it previously had. Command *0x64* turns off the LED again.

Commands *0x42* and *0x52* are used to cycle through an array of colors, either going left by decrementing the index with 0 going to 2, or by going right by incrementing the index modulo 3. The array it loops through, could potentially be extended to contain even more colors, making the looping more exciting and effective.

The main loop is continuously keeping track of an internal timer. This timer is a safety mechanism, to prevent the loop from executing the same command twice in quick succession. To compensate for inaccuracies in the aiming of the IR transmitter, the transmitter sends the same message 10 times in a row, to increase the chances of the receiver receiving the command. As a result, the LED can for instance cycle through several colors, after only one flick. This is unwanted behavior, so a timer was installed, where the same command will be ignored if it is received within a few seconds of the previous commands.

5 EVALUATION

5.1 Study Design

With the high-fidelity prototype, we conducted an evaluation study to understand both the user experience and the enchantment factor of our product. While usability evaluations often incorporate task performance metrics such as completion time or error rate, these were not the primary focus of our study. Given our interest in emotional user experience and enchantment rather than the efficiency of the prototype, we prioritized subjective measures that captured experiential aspects of the interaction.

To address the first research question – “How can controlling devices around the house with a novel remote control, the Magic Wand, be made user-friendly?” – we compared the high-fidelity prototype to the initial low-fidelity prototype. During the early design stages of the product, we collected User Engagement Scale (UES) [8] scores to understand aspects of user experience such as focused attention, perceived usability, aesthetic appeal, and reward factor. This scale was chosen because it provides a nuanced framework by focusing on various factors of user experience, but also aspects that relate to enchantment. Moreover, given that the aim of this research is to explore how interaction design can evoke enchantment while maintaining usability, it is important to rely on subjective measures that measure how users experience the interaction, rather than just how it functions. The UES includes the dimensions: **focused attention**; **perceived usability**; **aesthetic appeal** and **reward factor**, all of which reflect how users feel during and after the interaction. Perceived usability, in this context, refers to how usable the system feels to the user, rather than technical performance. Aesthetic appeal is particularly important for the design of the prototype,

Part	Purpose
Arduino Uno	Controlling computational logic.
RGB LED light	Indicating when a gesture has been recognized and behaving accordingly.
Infrared receiver	Receiving the incoming signals.

Table 4: *Formal list of Parts - Receiver*

Library	Purpose
IRremote .hpp	Used to receive infra-red signals.
Arduino.h	Access to standard definitions for working with an Arduino Microcontroller.

Table 5: *Formal list of Libraries - Receiver*

since visual features are important to induce a sense of enchantment. The factors focused attention and reward align closely with the characteristics of an enchanting experience. By collecting UES scores for both prototypes, we aimed to evaluate whether our design refinements led to improvements in user experience.

The second research question, *"How enchanting is it for users to control household devices with a novel remote control, the Magic Wand?"*, was addressed through an experimental setup. With comparative usability testing we compared the enchantment level of the high-fidelity prototype to a conventional, manually controlled lamp. A within-subjects design allowed each participant to experience both conditions, reducing variability due to individual differences. To measure the enchantment level of each condition we used the **Enchantment Adjective Checklist (EAC)**, a standardized scale to quantify enchantment [5]. As the only known study to quantify enchantment, this method was originally developed for paranormal tourism, but we have adopted this method for our study in interaction technology. By comparing EAC scores between the conditions, we evaluated whether our product provided a more enchanting experience than traditional methods to control devices.

In addition to these quantitative measures, we also collected qualitative data through open-ended survey questions. These were intended to gather general impressions, user feedback, and recommendations for future improvements to the prototype. As the main focus of the study was quantitative, the qualitative data served to support and contextualize the findings from the standardized scales. To facilitate this mixed-methods approach, the evaluation study had the following structure:

- Condition 1: high-fidelity prototype
 - Quantitative measures: UES and EAC;
 - Qualitative measures: open-ended questions about overall user experience.
- Condition 2: control (conventional lamp)
 - Quantitative measure: EAC.

5.2 Participants

Ten participants were recruited from the Master's program in Human-Computer Interaction at Utrecht University through convenience sampling. All participants were students aged between 20 and 30 years old, and included six male and four female participants. Ethical considerations were assessed using the Ethics and Privacy

Quick Scan from Utrecht University, which determined that the study posed minimal risk to participants. Informed consent was obtained from all participants prior to their involvement in the study.

5.3 Apparatus

The study was conducted within the university where participants were recruited. The main setup included (I) the high-fidelity prototype; (II) a manually controlled lamp, which served as the control condition; (III) two *Qualtrics* surveys for data collection, one for each condition.

It is important to note that during the study, some technical difficulties were encountered with the high-fidelity prototype. Specifically, the battery was depleted, requiring the device to be connected to a laptop for power, which impacted its wireless functionality. Additionally, the prototype was not in its final design. Certain aesthetic elements, such as the use of tape for temporary fixes, did not match the intended design. All participants were informed about these issues beforehand, ensuring they understood that the current prototype was not representative of the final product. These technical and aesthetic limitations may have influenced the participants' experience and their responses, although these factors were consistent across all participants.

5.4 Procedure

Each participant completed two conditions: (1) interacting with the prototype and (2) interacting with a conventional lamp. To compensate for potential order / novelty bias, half of the participants experienced the prototype condition first, followed by the conventional control, while the other half performed the tasks in the reverse order. After completing the task for each condition, their experiences were recorded via the associated *Qualtrics* survey. Each session lasted approximately 15 minutes, with the experiment conducted in a casual setting to encourage natural interactions. This approach ensured that all participants experienced both conditions under comparable circumstances. The full experimental procedure is provided in the attachments.

For data analysis, we examined both quantitative and qualitative data. Enchantment levels were calculated by summing the number of adjectives checked by participants for each condition. The total sum was then associated with a score based on the table in

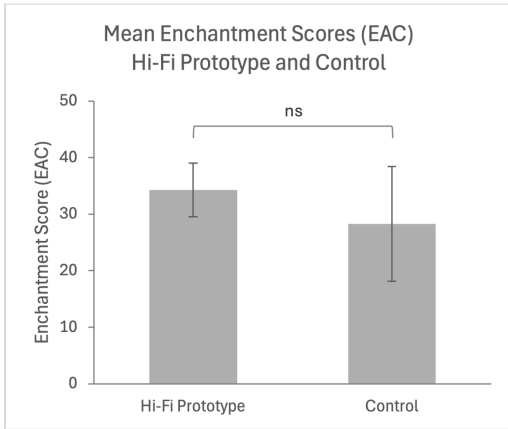


Figure 10: Mean EAC Score Comparison

[5], with higher sums indicating higher levels of reported enchantment. To compare the enchantment scores between the high-fidelity prototype and the control condition, a paired-samples t-test was conducted. For the UES, scores were calculated following the procedure outlined by [8]. First, reverse coding was applied to the items related to perceived usability. The scores for each of the four dimensions were calculated by adding the values of the three items in each subscale and dividing by three. An overall engagement score was calculated by adding the scores from all items and dividing by twelve. Finally, qualitative data from open-ended questions were summarized to highlight participants' opinions and feedback regarding their experience with the high-fidelity prototype.

6 RESULTS

6.1 EAC Scores

To evaluate enchantment levels during interaction we compared the high-fidelity prototype to a conventional manually controlled lamp using the Enchantment Adjective Checklist (EAC). While participants reported slightly higher enchantment levels for the high-fidelity prototype ($M = 34.3$, $SD = 4.74$) than for the control condition ($M = 28.3$, $SD = 10.12$), a paired-samples t-test revealed no significant difference ($t(9) = 1.837$, $p = 0.099$). This comparison is illustrated in figure 10.

Participants also rated their enchantment on a 4-point categorical scale ranging from *not at all enchanted* to *very enchanted*. In the prototype condition, 8 participants described themselves as *somewhat enchanted* and 2 as *very enchanted*. In contrast, responses in the control condition were more neutral: 5 participants reported being *barely enchanted*, and the other 5 reported being *somewhat enchanted*.

6.2 UES Scores

To examine how users experienced interaction with the prototype, we collected UES scores for both the low-fidelity and high-fidelity prototypes. However, direct comparisons are limited due to differences in the sample size and missing measures. The first iteration of the prototype was tested by 3 participants, and only two dimensions of the UES, perceived usability and focused attention, were

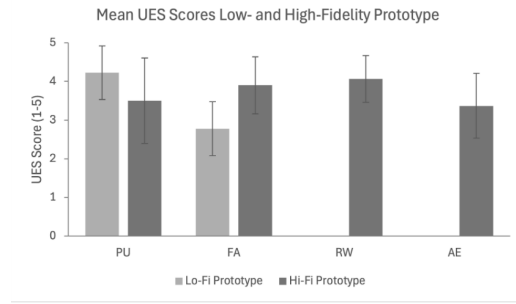


Figure 11: UES Score Comparison

measured. Conversely, the high-fidelity prototype was evaluated by all 10 participants and included all four UES dimensions. Therefore, no statistical comparisons can be made, and the results indicate trends rather than definitive conclusions.

For perceived usability (PU), the high-fidelity prototype scored lower ($M = 3.5$, $SD = 1.10$) than the low-fidelity prototype ($M = 4.22$, $SD = 0.69$). On the other hand, for focused attention (FA) the high-fidelity prototype scored higher ($M = 3.9$, $SD = 0.74$) compared to the low-fidelity prototype ($M = 2.78$, $SD = 0.69$). Furthermore, the high-fidelity prototype scored on aesthetic elements (AE) ($M = 3.37$, $SD = 0.84$) and reward factor (RW) ($M = 4.07$, $SD = 0.60$), which were not measured for the low-fidelity prototype. The average overall engagement score for the high-fidelity prototype was 3.71 ($SD = 0.52$), which indicates a moderately positive level of engagement according to the scale, with scores ranging from 1 to 5. This comparison is shown in figure 11

6.3 Qualitative Feedback

Open-ended responses offered further insight into the user experience. Many participants expressed enthusiasm for the concept and described their interaction as “fun”, “exciting”, and “satisfying”. In addition, some users mentioned feeling “in control”, while others noted a learning curve and indicated that it took a little time to get used to the gestures or understand how to interact with the system. One participant said: **“I felt like I was struggling, but when I was able to do it, I felt like I had mastered it.”**

Participants also offered suggestions for improvement, which fell into two categories: functionality and design. Seven of them commented on functional aspects of the prototype, suggesting improvements such as making gesture recognition more distinct, eliminating the need to physically point at the IR sensor, and adding clearer feedback through features like LEDs. One participant suggested highlighting the area where users should tap by using a different colour to improve intuitiveness, while another recommended removing the tapping requirement altogether. Furthermore, three comments about design-related aspects were made. One user stated that the distinction between the handle and the top part felt awkward, suggesting that the design could be more cohesive. In contrast, another participant mentioned that the handle should provide clearer guidance on how to hold the device. Finally, one person suggested making the device heavier to improve its physical feel.

7 DISCUSSION

7.1 Interpretation of Results

This study aimed to explore how interactive design can evoke feelings of enchantment while also considering usability aspects. The results offer preliminary insights into how users experience enchantment when interacting with a high-fidelity prototype, the *Magic Wand*. Although the prototype received higher enchantment scores than the control, this difference was not significant ($p > 0.05$). This suggests that while the concept shows potential, further refinement and testing are needed to evoke a strong sense of enchantment. However, self-reported enchantment on the categorical scale provided a more optimistic view, with at least eight participants describing themselves as “somewhat enchanted” when interacting with the prototype, compared to more neutral responses for the control. This inconsistency between measured scores and subjective reports suggests that enchantment may not be fully captured by quantitative metrics alone. Besides, it is also important to note that the high-fidelity prototype was neither wireless nor presented in its intended design form during testing, which may have influenced participants’ overall experience.

Secondly, responses to the UES indicated a generally positive score for the high-fidelity prototype, however, further testing with a larger sample size is necessary to draw more reliable comparisons between the prototypes. In terms of usability, the lower perceived usability score for the high-fidelity prototype compared to the low-fidelity version may reflect the impact of added complexity once interactive functionality is introduced. This was also reflected in participants’ qualitative comments, where several noted a learning curve when adapting to gesture-based interaction. Overall, the qualitative feedback supported the quantitative findings, with many participants describing the concept as exciting and the interaction as enjoyable. At the same time, they pointed out practical challenges related to gesture recognition and intuitiveness. These observations highlight important areas for improvement in both functionality and design.

7.2 Ethics Issues

The potential ethical issues associated with our project are minimal. This is primarily because we are modifying an existing device, such as a remote control, by adding an element of enchantment, like using spells instead of pressing buttons. Consequently, the ethical concerns are not so much about the enchantment itself, but rather about the general principles behind remote control devices.

Remote control devices, however, do raise some ethical and moral questions. For example, they can make it easier for individuals with malicious intent to control devices without being physically present. This can be particularly concerning when considering scenarios where, for instance, children might use generic IR remotes to turn off televisions in strangers’ homes. Fortunately, our product already addresses these issues. Instead of sending generic IR messages, we transmit codes that must be decrypted by a specific machine before they can control any device.

In our adversary design fiction, we illustrated how the *Magic Wand* could potentially disable any device, for example even breathalyzer devices in hospitals, which could have fatal consequences. In

practice, however, the signals sent by our *Magic Wand* are limited to IR messages with codes that only affect a predefined set of devices.

7.3 Social Impact

The social impact of the *Magic Wand* is limited to the global interest in making household routines more enchanting. While it does not change functionality, it may shift one’s view from efficiency to being more present in the moment, since doing a spell requires a level of engagement that a simple remote control button does not. The impact is not huge; it may simply bring some smiles to the faces of our users. It will not replace original remote control devices all over the world, as there will always be some people who simply prefer efficiency. If everyone were to use this device, it could slow many of us down while also encouraging us to be more present and appreciate the little things a bit more. One, perhaps far-fetched impact, is that people who are a bit nerdy and struggle with making conversation when hosting guests will now have an easy conversation starter or party trick.

7.4 Tangibility, Materiality and Embodiment

The *Magic Wand* is a tangible human interface device that users must hold to operate. It is designed with a specific grip to ensure proper orientation, as its sensors require a fixed, pre-determined starting position. To intuitively guide users, the handle features carved finger placements that promote a comfortable grip when held correctly. Incorrect grips result in discomfort, deliberately encouraging proper handling.

In terms of materiality, our initial intention was to craft the wand from wood, as user research indicated that a wooden appearance and texture evoke the strongest sense of enchantment. However, carving wood did not allow us to achieve the precise shapes we envisioned. As a result, we opted for 3D-printed material. To preserve the wand’s enchanting quality, we ensured that its appearance closely resembles wood. This includes using a wood-like color, incorporating carved details to simulate aged wood, and adding spiraling thorn-like textures. Through this approach, we believe we have successfully balanced manufacturability and enchantment in our material selection.

7.5 Limitations

Throughout our project, we encountered several challenges that affected the final device. Whether these should be considered limitations or mistakes is open to interpretation. The biggest challenge was recognizing more complex gestures, such as spinning motions. We dedicated a significant amount of development time to this, but were unable to achieve satisfying results. It turned out to be more difficult than expected.

Another issue became apparent later in the process. As we added more gestures, it became increasingly difficult to distinguish them reliably in code without triggering many false positives. This led to the removal of several features and gestures, which in turn resulted in a considerable loss of time and effort.

Finally, we originally planned to build the wand out of wood to enhance the magical aesthetic. However, due to limited equipment available at the lab, we could not find a practical way to make the device from wood. In the end, we believe the alternative we

found worked very well. The finalized 3D-printed material looks surprisingly close to wood, and we feel it gave us the best possible result given the tools we had.

7.6 Future Work

In the future, we would like to explore the possibility of implementing more complex, multi-gesture "spells." Unfortunately, this was not feasible within the limited timeframe of the project. Additionally, the current demonstration only included a receiving unit that controlled a basic RGB LED. Future work could extend our modular approach to integrate with home hubs such as Apple HomeKit or similar platforms, allowing control over a wider range of smart home devices.

Using Wi-Fi or Bluetooth instead of infrared could also significantly improve the system's stability and ease of use, as it would eliminate the need to point directly at the receiver. That said, several users mentioned that the pointing interaction felt intuitive.

8 CONCLUSION

The goal of the project was to develop a human-interface device that would make household routines more interactive and enchanting. To achieve this, we settled on the design of an innovative handheld *Magic Wand* that could toggle everyday household devices through the use of gestures.

Ultimately, we tried to answer two main research questions:

- (1) How can controlling devices around the house with a novel remote control – the *Magic Wand* – be made user-friendly?
- (2) How enchanting is it for users to control household devices with a novel remote control, the *Magic Wand*?

To answer the first question, we conducted user research to gather feedback on the device, which we incorporated into the development process. This iterative approach led to several improvements in the final design of the Magic Wand. A final user study was conducted involving participants who interacted with both an early low-fidelity prototype and a later high-fidelity version. By comparing User Engagement Scores (UES) between the two groups, we found that the low-fidelity version performed better in terms of perceived usability, while the high-fidelity version scored better on focused attention. However, drawing definitive conclusions was difficult due to differing sample sizes and the fact that not all UES components were measured for the low-fidelity version.

To address the second research question, we conducted a within-subject experimental study comparing the enchantment factor of our high-fidelity prototype to traditional methods of home device control. We used the Enchantment Adjective Checklist (EAC) to quantify and compare users' feelings of enchantment. Results showed that the Magic Wand did not provide a significantly higher level of enchantment during user interaction compared to a manually controlled lamp, although self-reported enchantment ratings suggested a positive user experience with the prototype.

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